Experiment 5

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Section/Group: KRG 3-A
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Subject Name: System Design
Subject Code: 23CSH-314

1. **Aim**: Design a Messenger Application(Similar to WhatsApp / Facebook Messenger): To design and analyze the architecture of a scalable real-time messenger application that supports one-to-one and group communication with high availability and low latency.

2. **Objective**:

- To implement real-time message delivery using WebSockets.
- To support one-to-one and group messaging.
- To preserve chat history.
- To ensure high availability and reliability.
- To design APIs for user management and chat operations.
- To design the system for large-scale usage (up to 1 Billion users).

3. **Tools Used:**

- **Frontend (ReactJS, HTML, CSS, JavaScript, WebSocket)** ○ Built responsive UI with real-time communication support
- **Backend (Node.js / Java / Spring Boot)** ○ Developed scalable REST APIs & WebSocket server
- **Database (MongoDB / MySQL / PostgreSQL)** ○ Managed structured & unstructured data efficiently
- **Redis** ○ Implemented caching & pub-sub for high performance
- **Deployment (AWS / GCP, Load Balancer, Docker)** ○ Cloud deployment with scalability, containerization & traffic distribution

4. **System Requirements:**

A. Functional Requirements

- **User Registration**
  a. POST /user/register
  b. Fields: userName, email, phoneNumber
- **User Login**
  a. POST /user/login

- **One-to-One Messaging**
  a. WS: /messages/send
  b. GET /messages/{userId}/{receiverId}
  c. Pagination support
- **Group Messaging**
  a. POST /groups/create
  b. POST /groups/{groupId}/add
  c. DELETE /groups/{groupId}/remove
  d. WS: /messages/send
- **Message Types**
  a. Text messages
  b. Media messages (images, videos, files)
- **Message History**
  a. Persistent storage
  b. Chat list view
- **Read Receipts**
  a. Delivered
  b. Seen

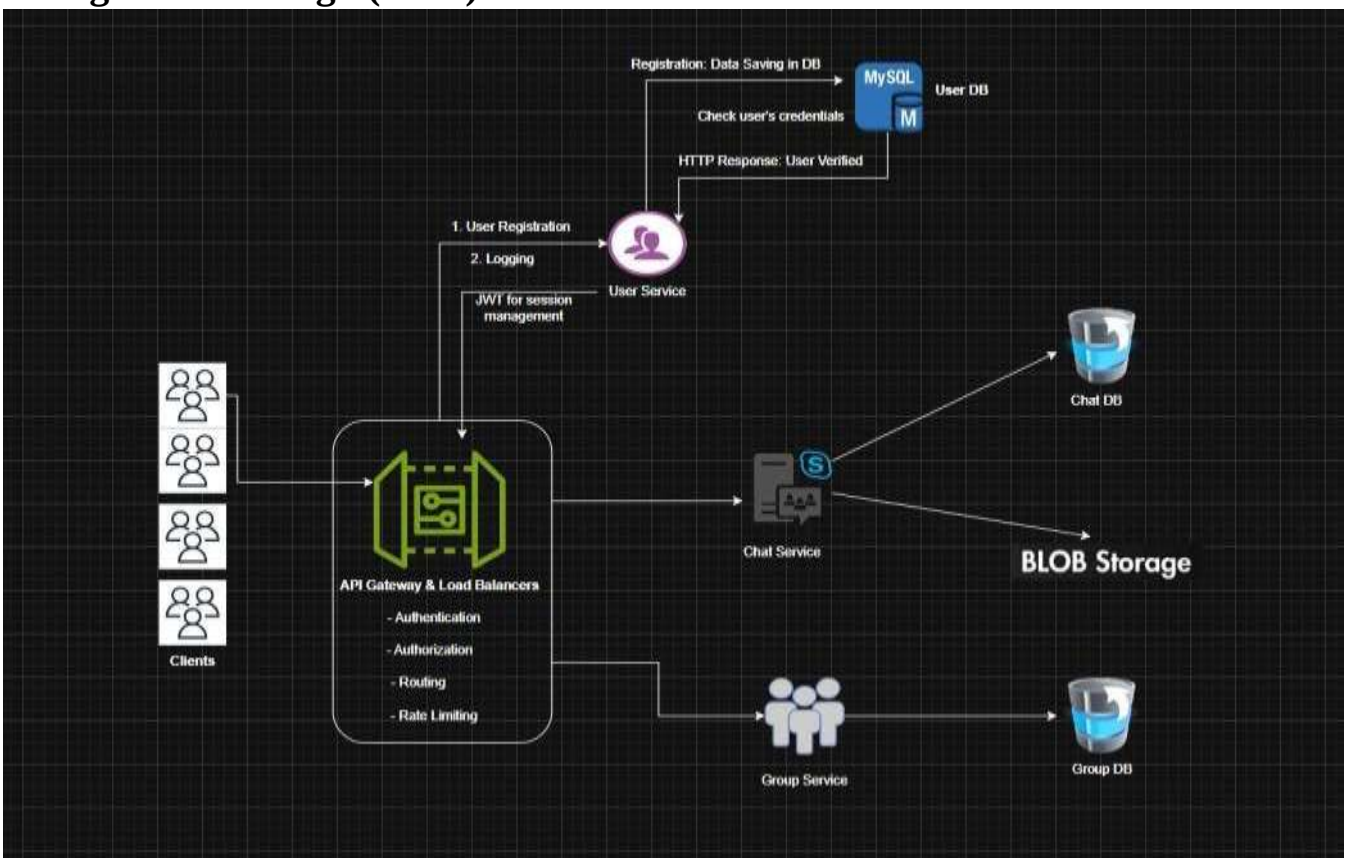
B. Non-Functional Requirements

- **Scalability**
  - Target: **1 Billion users**
  - 100 messages per user per day
  - ≈ 100 Billion messages/day
  - Estimated storage: ~100 TB

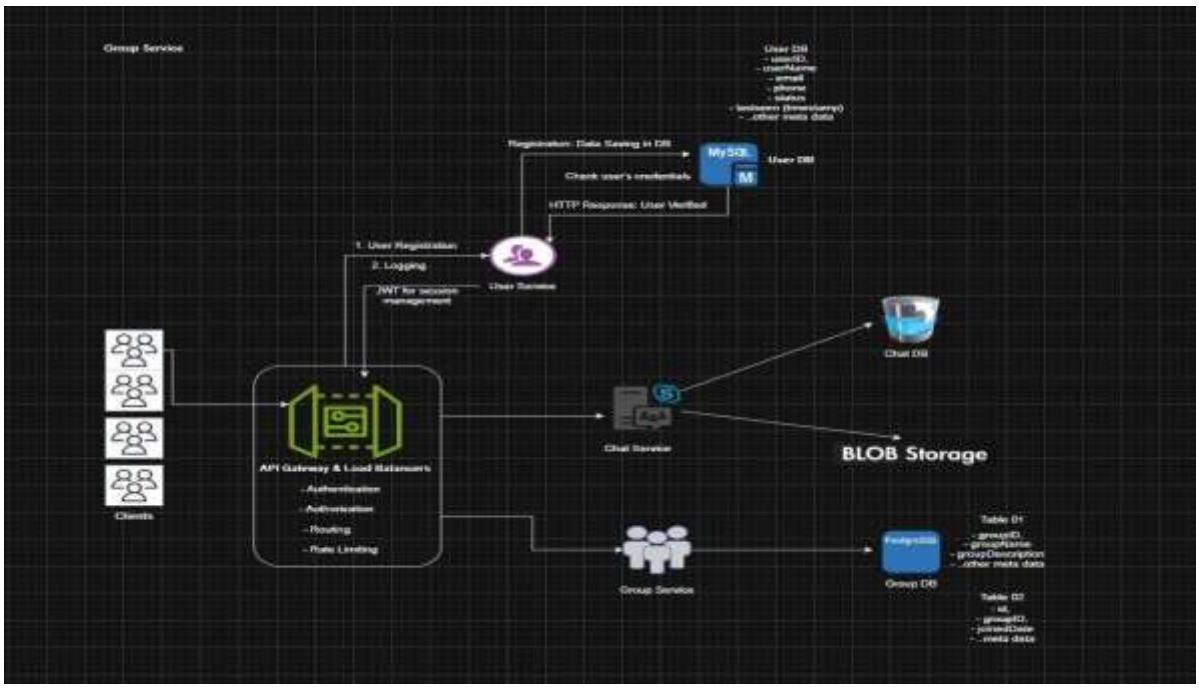
- **CAP Theorem**
  System prioritizes:
  - High Availability
  - Partition Tolerance
  - Eventual Consistency

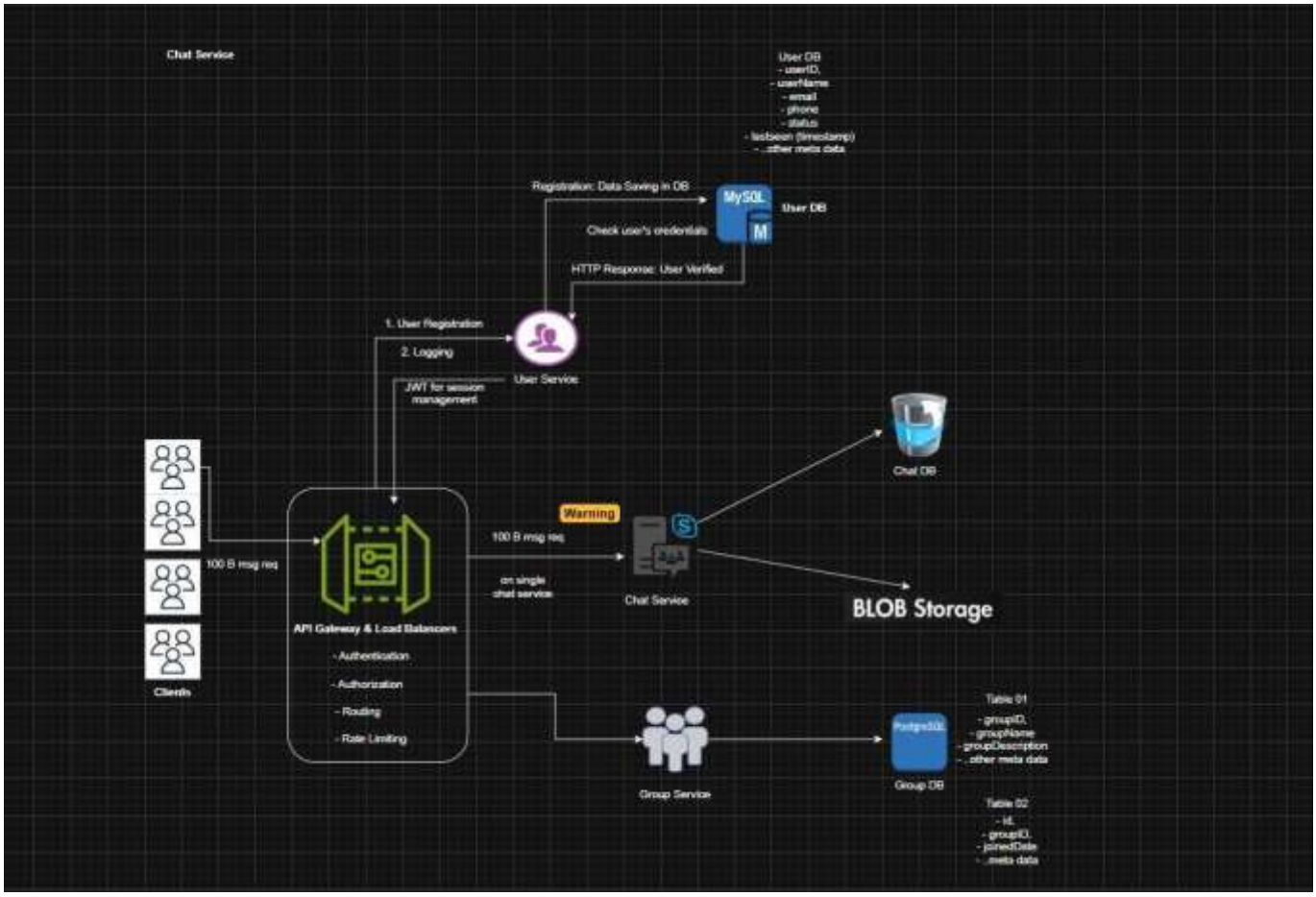
- **Latency: 200-300ms**
  - Target message delivery time: **200–300 ms**

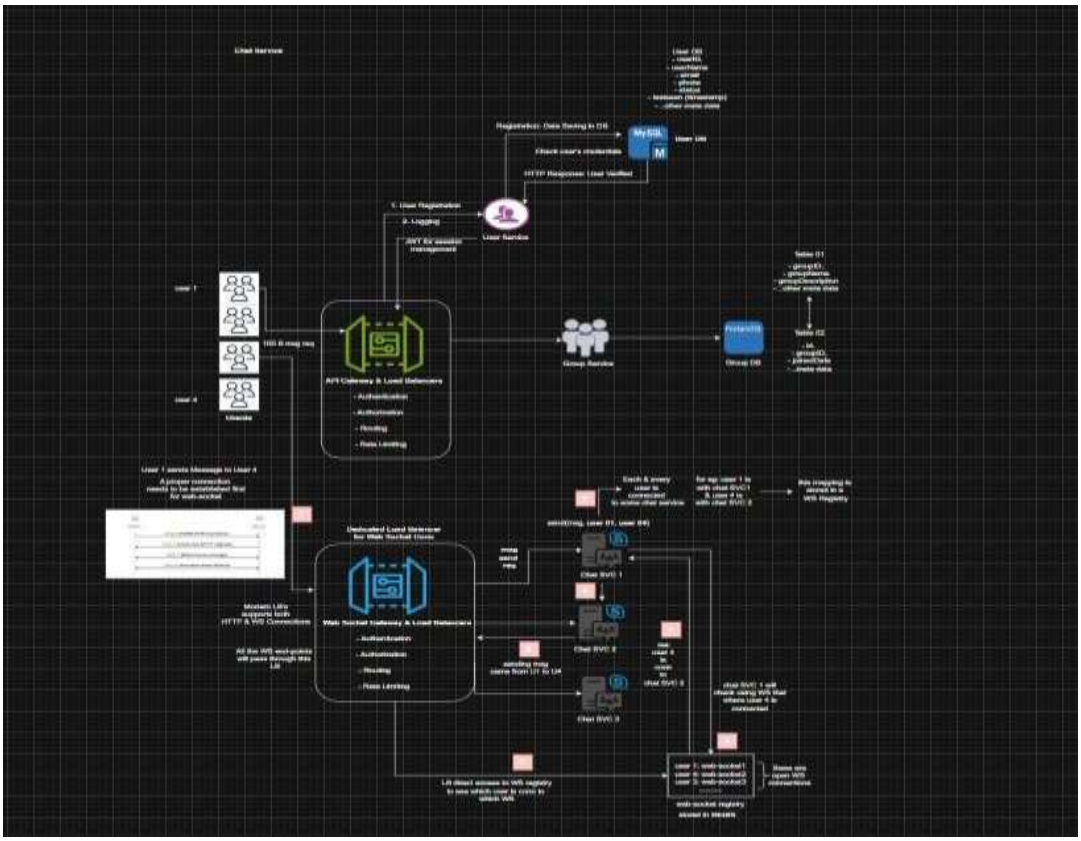
5. High Level Design (HLD)

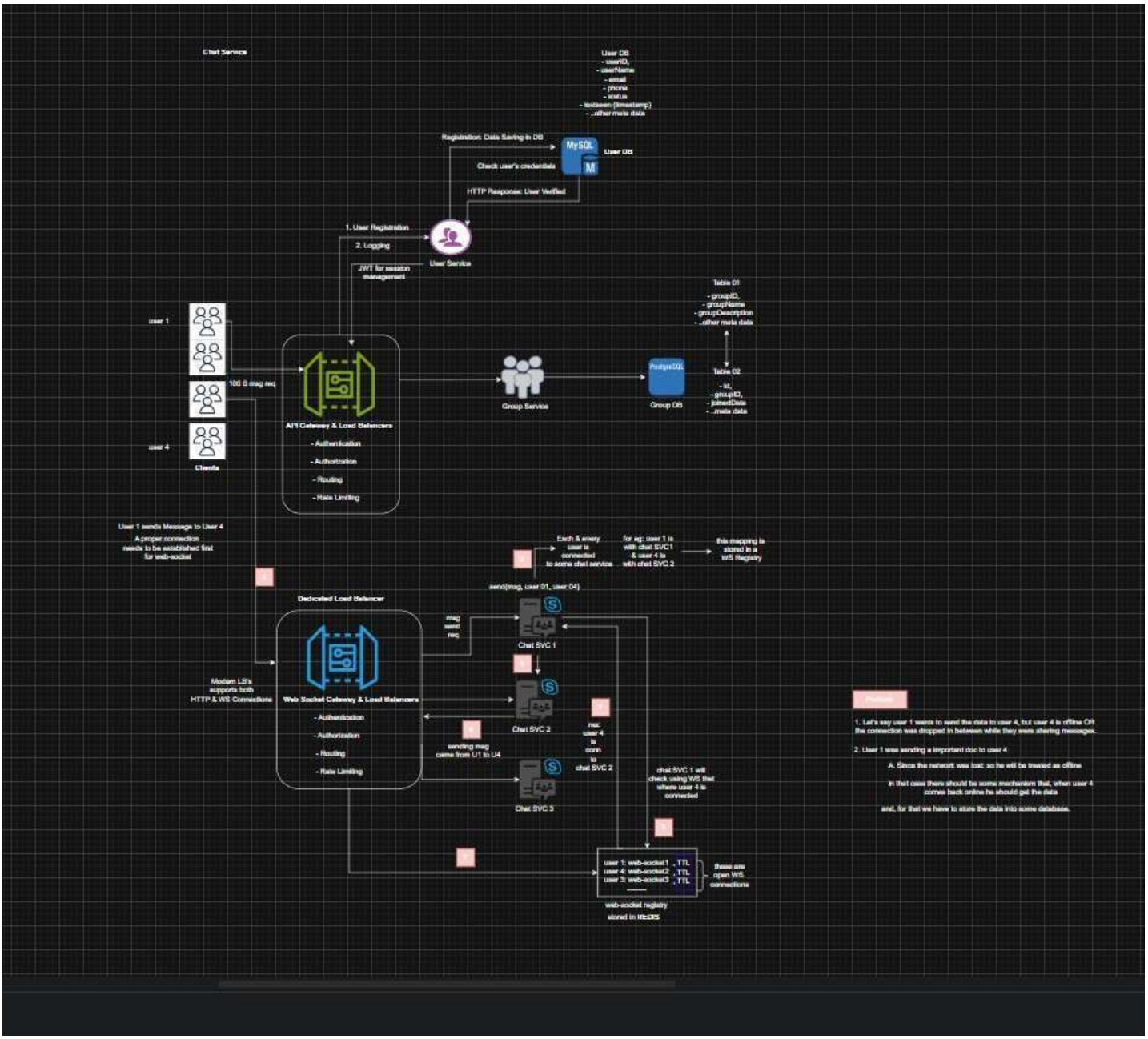


6. Low Level Design (LLD):




Chat Service
User DB
- userID,
- userName
- email
- phone
- status
- lastseen (timestamp)
- ...other meta data
Registration: Data Saving in DB
MySQL
User DB
Check user's credentials
HTTP Response: User Verified
1. User Registration
2. Logging
User Service
JWT for session management
Chat DB
Warning
100 B msg req
on single chat service
Chat Service
BLOB Storage
100 B msg req
API Gateway & Load Balancers
- Authentication
- Authorization
- Routing
- Rate Limiting
Clients
Group Service
Table 01
- groupID,
- groupName
- groupDescription
- ...other meta data
Group DB
Table 02
- id,
- groupID,
- joinedDate
- ...meta data

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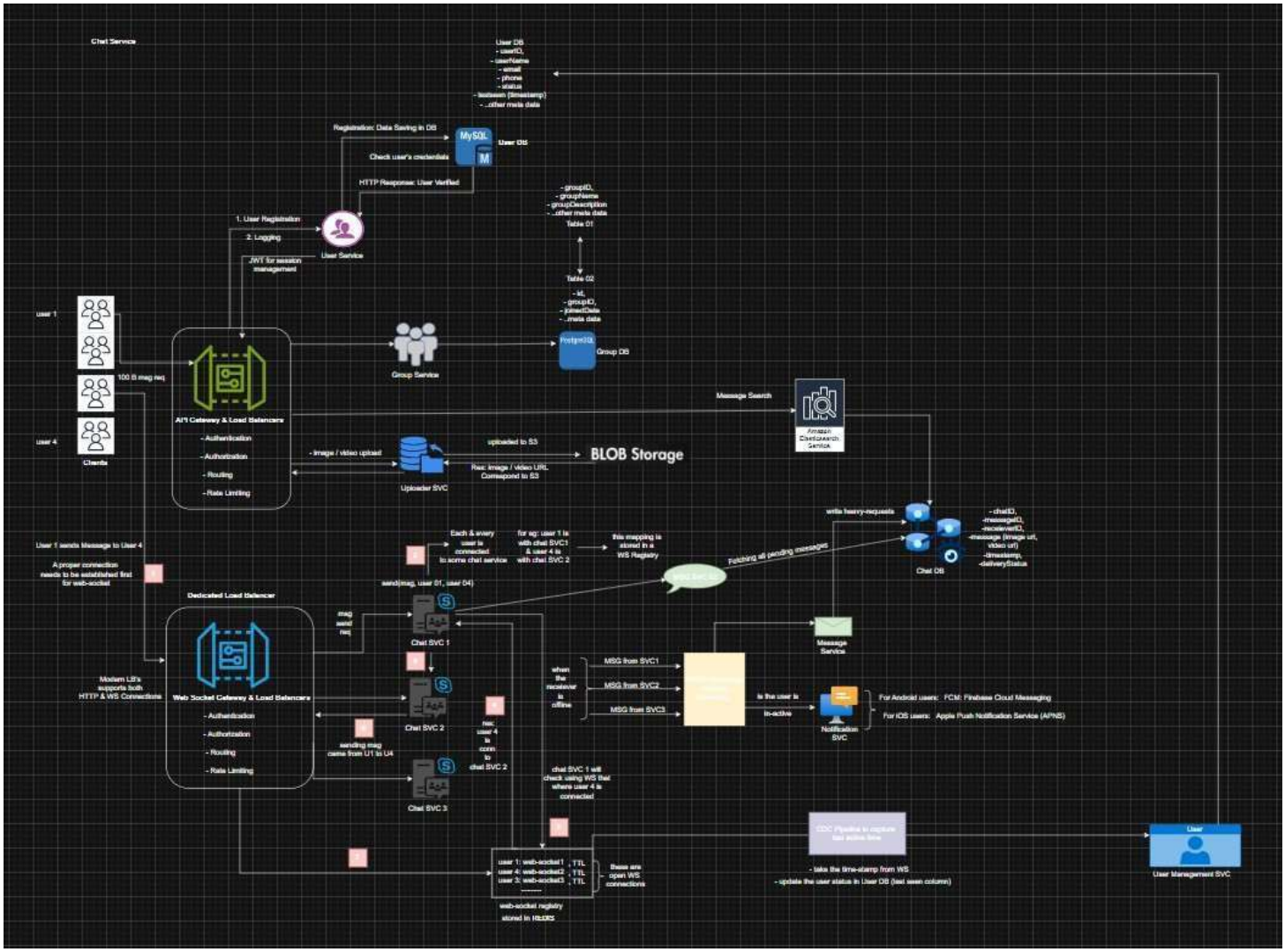
user 1
100 B msg req
user 4
Clients
API Gateway & Load Balancers
- Authentication
- Authorization
- Routing
- Rate Limiting
Group Service
PostgreSQL
Group DB
User 1 sends Message to User 4
A proper connection needs to be established first for web-socket
Each & every user is connected to some chat service
for eg: user 1 is with chat SVC 1 & user 4 is with chat SVC 2
this mapping is stored in a WS Registry
send(msg, user 01, user 04)
Dedicated Load Balancer
msg send req
Chat SVC 1
Modern LB's supports both HTTP & WS Connections
Web Socket Gateway & Load Balancers
- Authentication
- Authorization
- Routing
- Rate Limiting
Chat SVC 2
req: user 4 is conn to chat SVC 2
sending msg came from U1 to U4
Chat SVC 3
chat SVC 1 will check using WS that where user 4 is connected
user 1: web-socket1 , TTL
user 4: web-socket2 , TTL
user 3: web-socket3 , TTL
these are open WS connections
web-socket registry stored in REDIS
Problem
1. Let's say user 1 wants to send the data to user 4, but user 4 is offline OR the connection was dropped in between while they were sharing messages.
2. User 1 was sending a important doc to user 4
A. Since the network was lost so he will be treated as offline
In that case there should be some mechanism that, when user 4 comes back online he should get the data
and, for that we have to store the data into some database.



7. Scalability Solution

- **Horizontal Scaling**
  Add multiple application servers behind a load balancer to distribute traffic and handle billions of requests efficiently.
- **Database Sharding**
  Partition data based on userID or chatID so that message load is distributed across multiple database servers.
- **Caching with Redis**
  Store frequently accessed chats and recent messages in Redis to reduce database load and improve response time.
- **Message Queue System**
  Use Kafka/RabbitMQ for asynchronous message processing and reliable message delivery.
- **CDN for Media Files**
  Use Content Delivery Network to efficiently deliver images, videos, and files globally with low latency.

8. Learning Outcomes (What I Have Learnt)

- Understand real-time communication using WebSockets.
- Learn how to design scalable distributed systems.
- Apply CAP Theorem concepts in real-world system design.
- Design RESTful APIs and WebSocket endpoints.
- Analyze trade-offs between consistency, availability, and latency.

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